

Exploratory Data Analysis Report

Product Demand Forecasting Using Ridge and Lasso Regression

Course	Machine Learning
Dataset	Retail_Dataset.csv
Records Analyzed	169,211
Name	Kairav Vashi
Roll No	2520030582
Section	8

1. Introduction & Objective

This report presents the exploratory data analysis (EDA) conducted on Retail_Dataset.csv as part of the Product Demand Forecasting project. The objective of this analysis is to understand the structure, distribution, and relationships within the data before applying Ridge and Lasso Regression models. EDA helps surface data quality issues, informs preprocessing decisions, and provides early evidence for which features are likely to matter for prediction.

2. Dataset Overview

The dataset contains 169,211 records of historical daily product order demand across 4 warehouses and 30 product categories, covering 2,117 unique products. Each record captures the order demand for a specific product on a specific date, along with contextual features such as warehouse, promotion status, holiday indicators, and petrol price.

Column	Description
Product_Code / Product_id	Product identifiers
Warehouse	4 categories (A, C, J, S)
Product_Category	30 categories
Date	Daily record date
Order_Demand	Target variable
Open, Promo	Store status & promotion flags
StateHoliday, SchoolHoliday	Holiday indicators
Petrol_price	Daily fuel price index

3. Data Cleaning & Preprocessing Summary

Before conducting EDA, the dataset was checked and cleaned as follows:

- **Missing Values & Duplicates:** Checked explicitly — none found, in either full-row duplicates or duplicate (Product, Date) pairs.
- **Date Parsing:** The Date column was parsed as M/D/YYYY and used to extract Year, Month, Day, and DayOfWeek as separate numeric features.
- **Outlier Treatment:** Order_Demand is heavily right-skewed (max value of 2,500,000 vs. a 99th percentile of ~80,000). Extreme values were capped (Winsorized) at the 99th percentile so they would not dominate the regression loss function.
- **Categorical Encoding:** Warehouse, StateHoliday, and Product_Category were one-hot encoded.
- **Feature Scaling:** Petrol_price was standardized using StandardScaler, since Ridge and Lasso penalize coefficient magnitude and unscaled features would be penalized unfairly.

4. Exploratory Data Analysis

The following visualizations examine the distribution of the target variable, its relationship with categorical and contextual features, trends over time, and correlation among numeric features.

4.1 Distribution of Order Demand

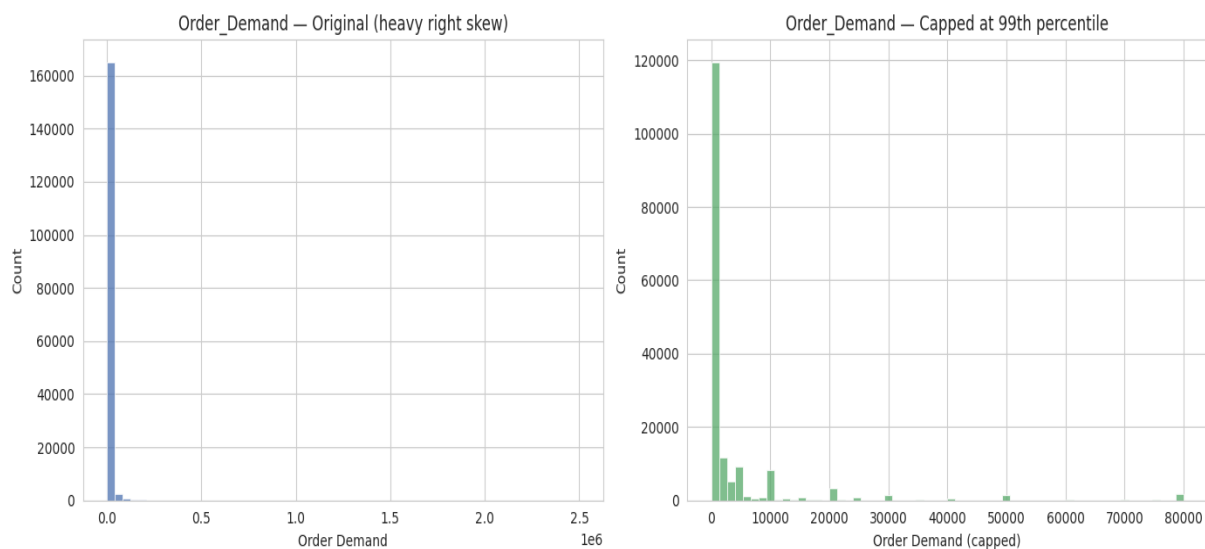


Figure 1: Order_Demand distribution before and after capping at the 99th percentile.

Key Observation

Order_Demand is extremely right-skewed — most orders are small, but rare bulk orders reach up to 2.5 million units. Left untreated, these extreme values would dominate the regression loss and distort model coefficients. Capping at the 99th percentile produces a distribution the model can learn from more effectively.

4.2 Demand by Warehouse

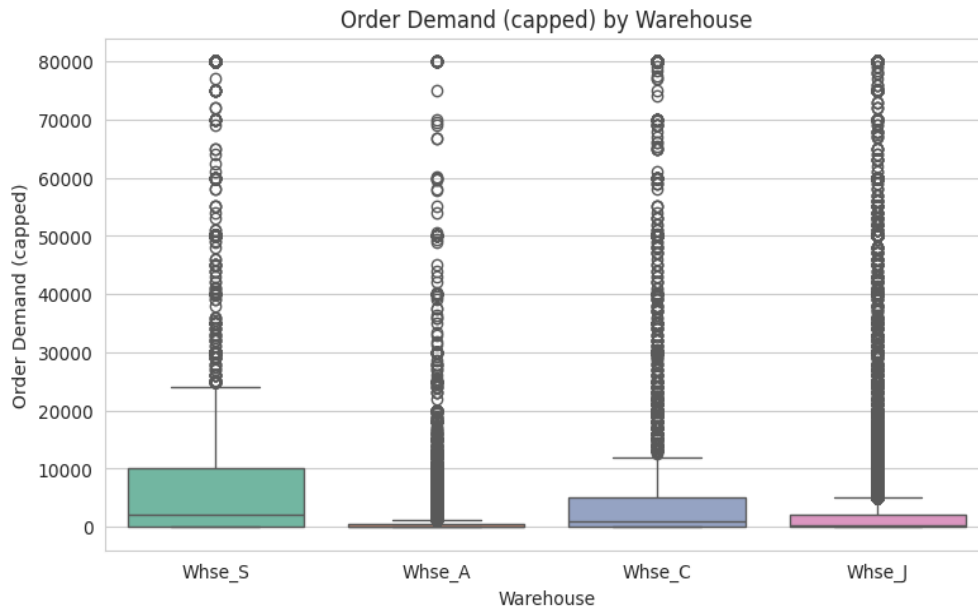


Figure 2: Boxplot of capped Order_Demand across the 4 warehouses.

Key Observation

Demand varies noticeably across warehouses — Whse_S and Whse_C show higher median demand than Whse_A. All warehouses show long tails of high-demand outliers, confirming the capping treatment was necessary hall-wide. Warehouse is likely a strong predictor, supporting its inclusion as a one-hot encoded feature.

4.3 Demand by Promotion and Holiday Flags

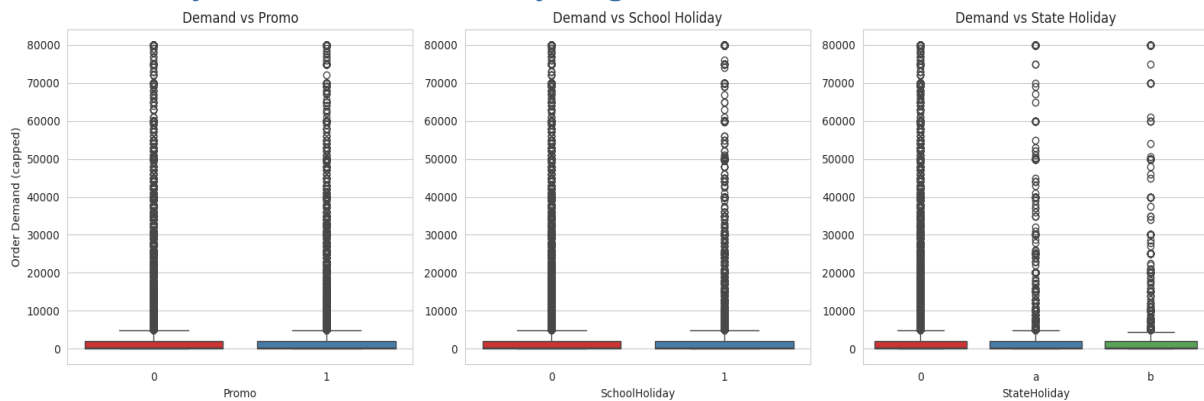


Figure 3: Demand distribution against Promo, School Holiday, and State Holiday flags.

Key Observation

Promo, SchoolHoliday, and StateHoliday show visible shifts in the demand distribution, though the effect appears smaller than the warehouse and category differences. These remain useful contextual features for the model even if individually weak.

4.4 Demand Trend Over Time

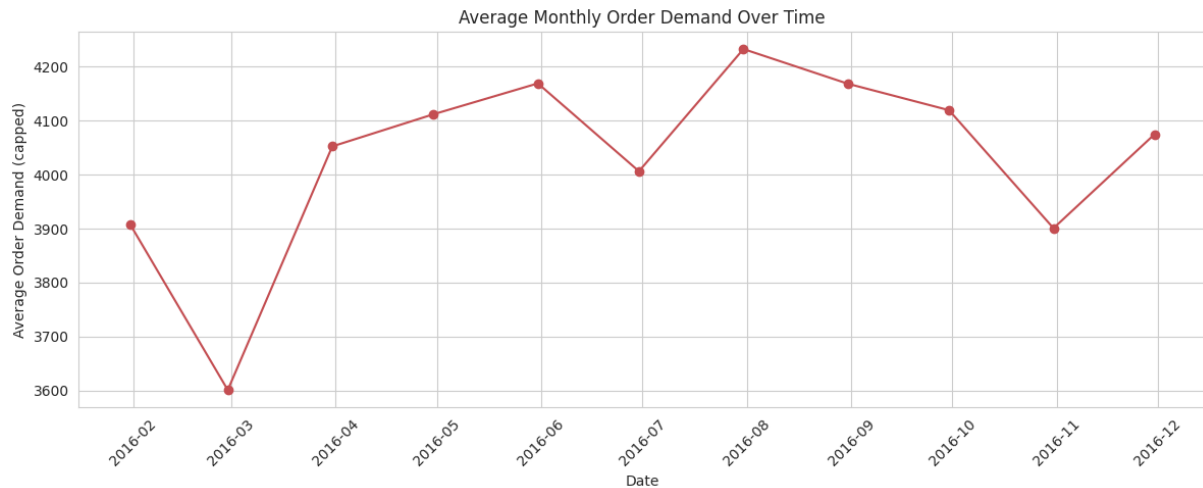


Figure 4: Average monthly order demand across the observed period.

Key Observation

Average monthly demand fluctuates moderately across the year without a strong long-term trend, suggesting demand is influenced more by product- and warehouse-level factors than by broad seasonal drift. Month was retained as a feature to capture any residual seasonality.

4.5 Demand by Product Category

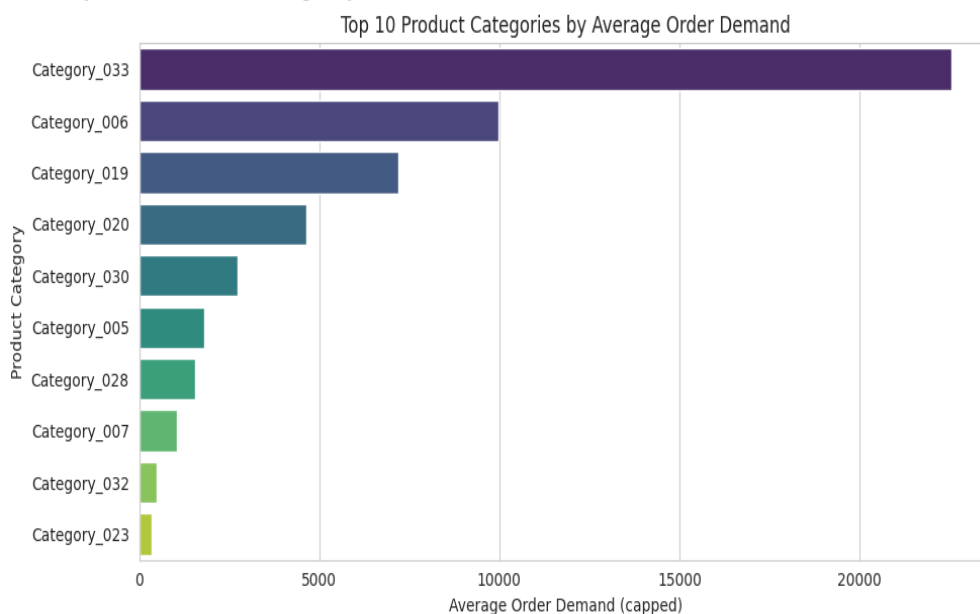


Figure 5: Top 10 product categories by average order demand.

Key Observation

Category_033 dominates average demand, more than double the next highest category. A steep drop-off after the top 3 categories suggests category-level effects are highly uneven rather than gradual. This reinforces Product_Category as a key feature, and hints that Lasso may zero out several low-impact category dummies during modeling.

4.6 Correlation Between Numeric Features

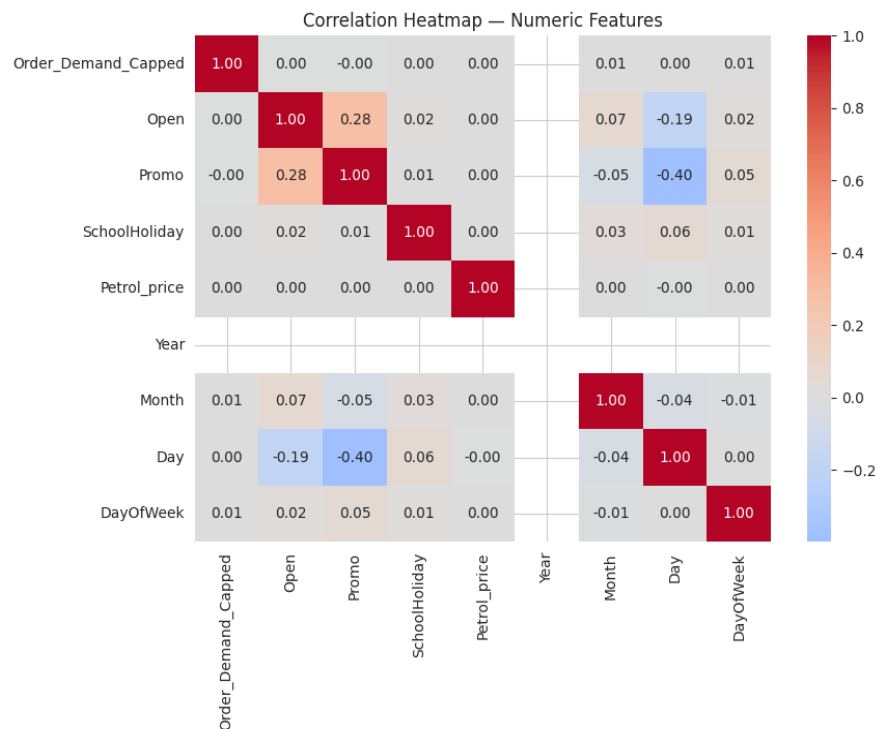


Figure 6: Correlation heatmap of numeric features against *Order_Demand*.

Key Observation

There is weak linear correlation between individual numeric features and demand — expected, since categorical features (Warehouse, Product_Category) likely carry more predictive signal. No strong multicollinearity is visible among these raw numeric features. However, once Warehouse and Product_Category are one-hot encoded (30+ new columns), multicollinearity risk rises across the full feature set — exactly the scenario Ridge and Lasso regression are designed to handle.

5. Summary of Key Findings

- Order_Demand is heavily right-skewed with extreme outliers, necessitating capping before modeling.
- Warehouse and Product_Category show the strongest visible relationship with demand among all available features.
- Promo and holiday flags have a smaller but noticeable effect on demand distribution.
- No strong monthly seasonality trend was observed, though minor fluctuations exist.
- Raw numeric features show weak individual correlation with demand, but one-hot encoding the categorical features will substantially raise the dimensionality and multicollinearity risk of the full feature set.
- This multicollinearity risk directly motivates the use of Ridge (L2) and Lasso (L1) regularization over plain linear regression for this project.

6. Conclusion

The exploratory analysis confirms that the dataset is suitable for a regularized regression approach. The heavy skew in the target variable, the strong categorical effects from Warehouse and Product_Category, and the multicollinearity expected after encoding all point toward Ridge and Lasso Regression as appropriate modeling choices. These findings directly inform the feature engineering and model training steps that follow in this project.